



Harshitha S 250701234-2025-CSE ▾

H2**Started on** Monday, 5 October 2026, 3:49 PM**State** Finished**Completed on** Monday, 5 October 2026, 3:50 PM**Time taken** 1 min 37 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     // Phase 1: Find intersection point
14     int slow = a[0];
15     int fast = a[0];
16
17     do {
18         slow = a[slow];
19         fast = a[a[fast]];
20     } while (slow != fast);
21
22     // Phase 2: Find entrance of the cycle
23     slow = a[0];
24
25     while (slow != fast) {
26         slow = a[slow];
27         fast = a[fast];
28     }
29
30     printf("%d", slow);
31
32     return 0;
33 }
34

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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